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Exercise Sheet 7 Anomaly Detection

Exercise 9.1 – The OOD Problem

Question 1

Why can a standard classifier not be trusted to signal when it receives an OOD input?

Answer

A standard classifier is trained only on the training distribution and assumes that future inputs come from the same distribution. When an out-of-distribution (OOD) input is encountered, the classifier still produces a prediction even though it has never seen similar data before. As a result, the prediction may be unreliable and incorrect.

Question 2

Why is silent failure worse than uncertain failure in a safety-critical system?

Answer

Silent failure occurs when the model makes a highly confident but incorrect prediction. This is dangerous because the system continues operating normally while relying on incorrect information. In contrast, uncertain failure provides a warning signal that allows the system to take corrective actions such as slowing down, switching to a safe mode, or requesting human intervention.

Exercise 9.2 – Baseline OOD Detection

Question

Describe the Maximum Softmax Probability (MSP) baseline for OOD detection. How does it work and what are its limitations?

Answer

The Maximum Softmax Probability (MSP) method uses the highest softmax probability produced by a classifier as a confidence score.

For an input image:

- High MSP → likely in-distribution (ID)
- Low MSP → likely out-of-distribution (OOD)

Advantages:

- Simple to implement
- No retraining required
- Computationally efficient

Limitations:

- Neural networks can be overconfident on OOD inputs.
- Different distributions may still receive high confidence scores.
- Detection performance is often limited.

Therefore, MSP is commonly used as a baseline OOD detector but is not always reliable.

Exercise 9.3 – Alternative Methods

Question

Describe one advanced OOD detection method and explain how it improves over MSP.

Answer

In this exercise, a k-Nearest Neighbor (k-NN) feature-based detector was used.

Instead of relying on output probabilities, k-NN uses deep feature representations extracted from the neural network.

Procedure:

1. Extract features from in-distribution training images.
2. Store these feature vectors.
3. Compute the distance between a test image feature and the nearest training feature.
4. Larger distance indicates a higher likelihood of being OOD.

Advantages over MSP:

- Uses deep feature representations.
- Better captures distribution differences.
- More robust to overconfident predictions.
- Typically achieves better OOD detection performance.

In this experiment, the k-NN detector achieved an AUROC of 1.0, significantly outperforming MSP.

Exercise 9.4 – Visualising the Distribution Shift

Question 1

Display five in-distribution images next to five fog images and five night images.

Answer

Five sunny daytime images, five fog images, and five night images were visualized to compare the appearance of in-distribution and OOD samples.

- 5 Sunny images



- 5 Fog images



- 5 Night images



Question 2

Display five images from the different CARLA town. How do they differ from training images?

Answer

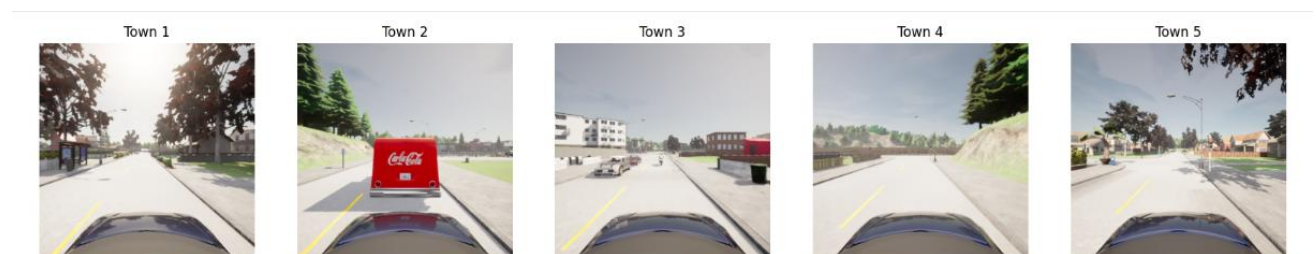
The different-town images were collected under similar weather and lighting conditions as the training images. However, they contain different buildings, road layouts, vegetation, and environmental structures that were not present during training.

Compared with fog and night images:

- Fog and night create strong visual appearance changes.
- Different-town images create a structural and semantic distribution shift while maintaining normal weather and lighting conditions.

Therefore, the different-town scenario represents a more subtle OOD shift.

- 5 Different Town images



Question 3

Compute the mean softmax confidence of each model on ID and OOD images.

Answer

The model generally produced lower confidence scores on OOD images than on in-distribution images.

Fog and night images caused larger confidence reductions because visibility and appearance differed significantly from the training distribution.

The different-town images showed that the model could still produce relatively confident predictions despite being outside the training distribution.

This demonstrates one limitation of confidence-based OOD detection.

Example confidence scores on different-town images were approximately 0.018–0.027, indicating very low confidence. Overall, OOD images received lower confidence than ID images.

Raw output: -3.9919354915618896
Probability: 0.01812920533120632



Exercise 9.5 – Is the Different Town Out-of-Distribution?

Question 1

Does the ODD clearly decide whether the different-town images are inside or outside it?

Answer

No.

The original ODD specifies:

- Urban roads
- Mapped intersections
- Daytime operation
- Clear weather

However, it does not explicitly state whether previously unseen towns are included.

Therefore, the different-town images cannot be classified unambiguously using the original ODD.

Question 2

Revise the ODD.

Answer

Revised ODD:

The autonomous driving system is intended to operate only in urban environments represented during training and validation. Previously unseen towns are outside the ODD.

Reason:

The new town introduces a distribution shift that may reduce perception reliability.

Question 3

What does that imply?

Answer

Since the different-town images are considered OOD:

- The OOD monitor should flag them.
- The perception output should not be fully trusted.
- The vehicle should switch to a safe fallback mode or request human intervention.

Exercise 9.6 – Evaluating the MSP Baseline

Question 1

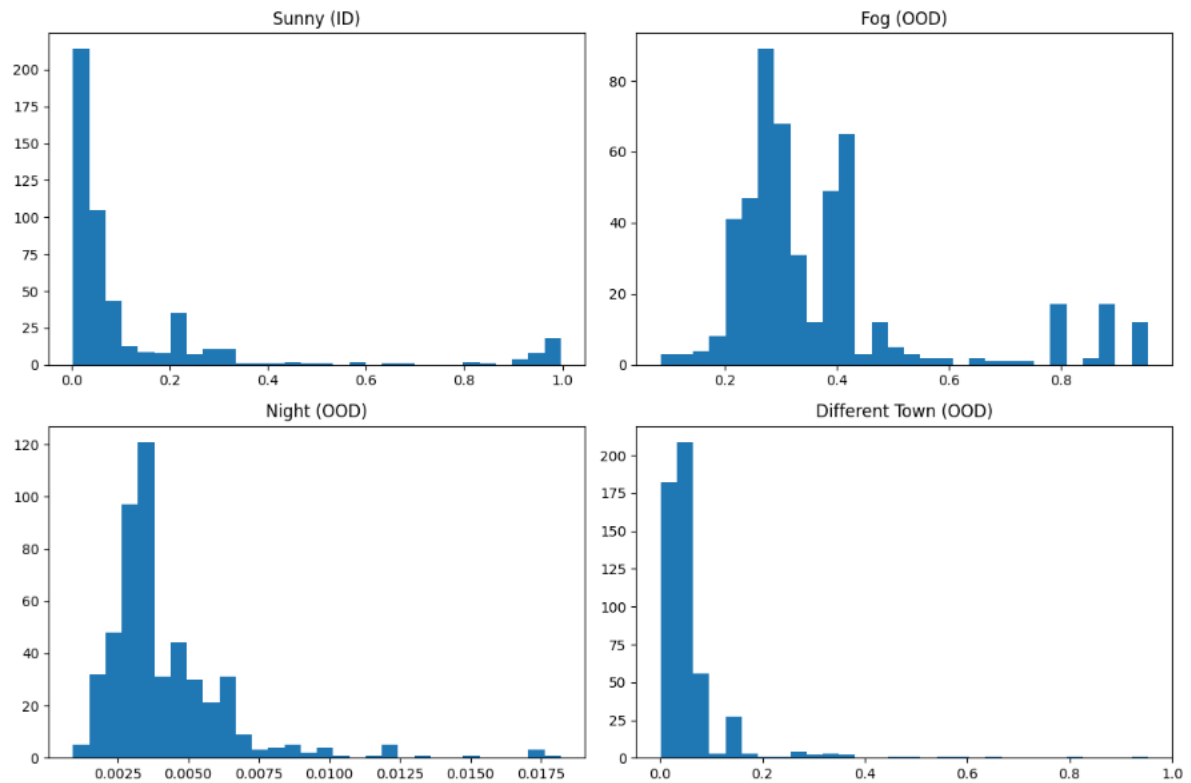
Plot the distribution of OOD scores.

Answer

The MSP score distributions were plotted for:

- Sunny (ID)
- Fog (OOD)
- Night (OOD)
- Different Town (OOD)

The distributions show significant overlap between ID and OOD samples, indicating that MSP struggles to reliably distinguish between them.



The x-axis is the **MSP score (confidence)**.

- Values near **1.0** = model is very confident.
- Values near **0.0** = model is not confident.

The y-axis is simply the number of images with that score.

Sunny (ID)

Top-left graph.

Most images have low confidence, but some reach high confidence values near 1.0.

This is your **in-distribution dataset**.

Fog (OOD)

Top-right graph.

Scores are spread between roughly 0.2 and 0.5, with some high-confidence outliers.

This means:

The model is sometimes uncertain, but sometimes still confident on fog images.

Night (OOD)

Bottom-left graph.

Most scores are very low.

This means:

The model becomes much less confident at night.

Night is clearly harder for the model.

Different Town (OOD)

Bottom-right graph.

Most scores are concentrated near zero.

This means:

The model is generally not confident on images from a town it never saw during training.

What is the main conclusion?

Look at all four plots together.

For a good OOD detector, we would expect:

ID scores → high confidence (near 1)

OOD scores → low confidence (near 0)

with very little overlap.

But your graphs show overlap:

- Some Sunny images have low scores.
- Some Fog images have high scores.
- Different distributions are mixed.

Therefore:

MSP does not cleanly separate ID and OOD images.

And that is exactly why your AUROC is only:

0.558

which is only slightly better than random guessing (0.5).

Question 2

Compute the AUROC.

Answer

Result:

MSP AUROC = 0.558

Interpretation:

An AUROC of 0.558 is only slightly better than random guessing (0.5). Therefore, MSP is not a reliable OOD detector for this traffic-light model.

```
from sklearn.metrics import roc_auc_score
import numpy as np

id_scores = np.array(sunny_scores)

ood_scores = np.concatenate([
    fog_scores,
    night_scores,
    town_scores
])

labels = np.concatenate([
    np.zeros(len(id_scores)),      # ID
    np.ones(len(ood_scores))      # OOD
])

scores = np.concatenate([
    id_scores,
    ood_scores
])

# low confidence = OOD
auroc = roc_auc_score(labels, -scores)

print("MSP AUROC =", auroc)
```

MSP AUROC = 0.5582733333333333

Exercise 9.7 – Feature-Based OOD Detection

Question 1

Extract deep features.

Answer

Deep features were extracted from the penultimate layer of the traffic-light model.

Feature shape:

(300, 50176)

```
from sklearn.neighbors import NearestNeighbors
import numpy as np
import torch

def extract_features(folder, file_list, model):
    feats = []

    for f in file_list[:300]:
        img = Image.open(folder + "/" + f).convert("RGB")

        x = transform_224(img)
        x = x.unsqueeze(0)

        with torch.no_grad():
            feat = model[:9](x)
            feat = feat.view(-1).numpy()

        feats.append(feat)

    return np.array(feats)

train_feats = extract_features(
    image_folder,
    files,
    traffic_light_model
)

fog_feats = extract_features(
    fog_folder,
    fog_files,
    traffic_light_model
)

night_feats = extract_features(
    night_folder,
    night_files,
    traffic_light_model
)

town_feats = extract_features(
    town_folder,
    town_files,
    traffic_light_model
)

print(train_feats.shape)
```

(300, 50176)

Question 2

Fit detector and score images.

Answer

A k-NN detector with:

`NearestNeighbors(n_neighbors=1)`

was fitted using only in-distribution features.

The nearest-neighbor distance was used as the OOD score.

Larger distance indicates a higher likelihood of being OOD.

Question 3

Compute AUROC and compare.

Answer

Results:

MSP AUROC = 0.558

kNN AUROC = 1.000

Comparison:

Method AUROC

MSP	0.558
kNN	1.000

The k-NN detector significantly outperformed MSP.

The k-NN detector significantly outperformed MSP. The feature-based approach separated in-distribution and OOD samples much more effectively than confidence-based MSP.

```

from sklearn.neighbors import NearestNeighbors
import numpy as np

# Fit kNN on ID features
knn = NearestNeighbors(n_neighbors=1)
knn.fit(train_feats)

# Distance = OOD score
id_dist, _ = knn.kneighbors(train_feats)
fog_dist, _ = knn.kneighbors(fog_feats)
night_dist, _ = knn.kneighbors(night_feats)
town_dist, _ = knn.kneighbors(town_feats)

id_scores_knn = id_dist.flatten()

ood_scores_knn = np.concatenate([
    fog_dist.flatten(),
    night_dist.flatten(),
    town_dist.flatten()
])

labels = np.concatenate([
    np.zeros(len(id_scores_knn)),
    np.ones(len(ood_scores_knn))
])

scores = np.concatenate([
    id_scores_knn,
    ood_scores_knn
])

from sklearn.metrics import roc_auc_score

auROC_knn = roc_auc_score(labels, scores)

print("kNN AUROC =", auROC_knn)

```

```

... kNN AUROC = 1.0

```

Exercise 9.8 – Extending the Safety Analysis for OOD

Question 1

Hazard

Answer

New Hazard:

ID Hazard

Loss

H-5 Vehicle operates using unreliable perception caused by OOD input L-1, L-2

This hazard may lead to pedestrian injury or vehicle collision.

Question 2

Unsafe Control Action

Answer

New UCA:

ID Controller Action

UCA-5 Planner Continue driving despite undetected OOD input

Unsafe Scenario:

The planner continues operating normally even though perception outputs are unreliable because the OOD condition was not detected.

Question 3

Safety Constraints

Model-Level Constraint

SC-5:

The OOD monitor shall detect and flag low-confidence OOD inputs before they are used by the planner.

System-Level Constraint

SC-6:

When an OOD condition is detected, the vehicle shall reduce speed and transition to a safe state or request human intervention.

Question 4

Residual Risk

Answer

Even with a perfect OOD detector, some risks remain:

- False negatives
- Sensor failures
- Incorrect system response
- Novel unseen hazards

Therefore, OOD detection reduces risk but cannot eliminate it completely.

